

Larger Type E Constructions for Queen Domination

Abstract

The queen domination problem asks for the minimum number of queens required to dominate an $n \times n$ chessboard. For boards of order $4k + 1$, Burger and Mynhardt showed that the existence of certain structured dominating sets of cardinality $2k + 1$, called Type E (i, i) - or $(i, i + 2)$ -sets, yields asymptotic upper bounds on the domination number whose coefficient approaches $1/2$ as k increases. Finding such sets for larger values of k is therefore of interest. We present a computational method for constructing members of a centrally symmetric subfamily of Type E (i, i) -sets. The construction problem is reformulated as a pairing problem on diagonal labels. By separating the labels according to parity and halving them, several choices that would otherwise be made explicitly during the search can instead be omitted and recovered deterministically afterwards. Necessary arithmetic conditions are used to restrict the parameter choices, and the resulting pairing problem is solved by constraint programming with randomized restarts. Using this method, we obtained nineteen verified constructions, the largest for $k = 396$ on Q_{1585} . Since $\gamma(Q_{4k+1}) \geq 2k + 1$, each such construction also determines the domination number exactly. The constructions therefore give $\gamma(Q_n) = (n + 1)/2$ for nineteen new values of n between 249 and 1585, whereas the largest value of n for which a set of this kind was previously known is 221. Through the Burger–Mynhardt bound the largest construction further gives $\gamma(Q_n) \leq \frac{1193}{2379}n + \mathcal{O}(1) \approx 0.5014712n + \mathcal{O}(1)$ for all sufficiently large n , improving the previous best known asymptotic upper bound.

Keywords: queen domination, queens graph, domination number, constraint programming, computational combinatorics

1 Introduction

We follow the notation of [1]. The queens graph Q_n has the squares of the $n \times n$ chessboard as its vertices, two squares being adjacent if they lie in a common row, column or diagonal. A set D of squares is *dominating* if every square is in D or adjacent to a square of D , and $\gamma(Q_n)$ denotes the minimum cardinality of a dominating set.

Raghavan and Venkatesan [2] proved $\gamma(Q_n) \geq \frac{1}{2}(n - 1)$, and Weakley [3] refined this to $\gamma(Q_{4k+1}) \geq 2k + 1$; see also [4] for a further sharpening for general n . Finding upper bounds that approach this theoretical lower bound is of interest. Burger and Mynhardt [1, Corollary 11] established the following: *if there exists a dominating (i, i) - or $(i, i + 2)$ -set of Q_{4k+1} of cardinality $2k + 1$, with $i < k$, then for all n large enough*

$$\gamma(Q_n) \leq \frac{3k + 5}{6k + 3}n + \mathcal{O}(1). \quad (1)$$

The coefficient $(3k + 5)/(6k + 3)$ is strictly decreasing in k and approaches $1/2$. The bound (1) is therefore a machine that converts a single dominating set on one board Q_{4k+1} into an asymptotic bound for all large n , and the quality of the bound is governed entirely by the largest k for which such a set is known.

A dominating set of Q_{4k+1} of cardinality $2k + 1$ has a second consequence that is independent of (1). Together with Weakley’s lower bound $\gamma(Q_{4k+1}) \geq 2k + 1$ it determines the domination

number exactly:

$$\gamma(Q_{4k+1}) = 2k + 1 = \frac{n+1}{2}, \quad n = 4k + 1. \quad (2)$$

Each construction of the kind sought here therefore contributes a new exact value of γ , as well as an instance of (1).

At the time of Burger and Mynhardt [1], the largest known Type E (i, i) -set of the required cardinality had $k = 28$, while the largest $(i, i + 2)$ -set had $k = 32$, corresponding to Q_{129} [5]; the latter gave $\gamma(Q_n) \leq \frac{101}{195}n + \mathcal{O}(1)$, and $\frac{101}{195} \approx 0.51795$. Neuhaus [6, Appendix B] later constructed sets that, in the terminology of [1], are (i, i) -sets for $k = 32, 39, 41$, and 46, and an $(i, i + 2)$ -set for $k = 55$ on Q_{221} , which yields the coefficient $170/333 \approx 0.51051$ through (1). The best published asymptotic coefficients, however, have come from a different class of constructions. Using Östergård and Weakley’s theorem for Type A covers containing no edge square [5, Theorem 5], Neuhaus [6, Cor. 3.24] obtained $\gamma(Q_n) \leq \frac{30}{59}n + \mathcal{O}(1)$, and $\frac{30}{59} \approx 0.5084746$. Kong subsequently reported a Type A construction yielding $17/33$ [7], and later a border-free construction yielding $112/221 \approx 0.5067873$ [8]; the latter improves Neuhaus’s coefficient. Other structured queen-domination constructions and related problems have also been studied, often using substantial computational search [5–13].

Despite these advances, the published structured constructions leading to such asymptotic bounds have remained on boards of order in the low hundreds. This suggests that a principal obstacle to further improving the bounds is not necessarily the absence of suitable theoretical results, but the difficulty of finding configurations satisfying their required structure on larger boards.

This paper develops a computational method for finding Type E (i, i) -sets. The class of configurations is not new: it is the family introduced in [1] and catalogued in [6], restricted here to its centrally symmetric members. What differs is the representation used for the search. Rather than searching directly over queen placements or prescribed line assignments, we describe an orientation-free halved-label pairing representation in which some of the choices are removed from the search altogether and recovered deterministically afterwards. This formulation and the class of configurations considered are developed in Section 2; the pairing problem is encoded as a feasibility problem and solved with CP-SAT using randomized restarts in Section 3; and the effect of the representation on performance is examined in Section 5.

Relation to previous computational work. The two-stage architecture used here—deriving necessary arithmetic conditions on a small set of parameters, enumerating the admissible parameter tuples, and then solving a combinatorial feasibility problem for each—is that of Östergård and Weakley [5], who determined the admissible parameter tuples from congruence and inequality conditions and then solved each resulting instance as an exact-cover problem using Knuth’s dancing-links algorithm. What differs here is the representation in which the second stage is posed, and the use of randomized restarts, whose relevance for this problem family was indicated by the search-time variation reported in [6]. The present work is unrelated in method and in scope to [13], which concerns small boards ($n \leq 19$): that work enumerates all minimum dominating sets up to isomorphism by proof-producing SAT solving on the full $n \times n$ board, and uses none of the Type E structure, reduced-board compression, parameter filtering or centrally symmetric restriction employed here.

Using this method, Type E (i, i) -sets were obtained for several larger values of k , the largest being $k = 396$ on Q_{1585} . By (2) these determine $\gamma(Q_n)$ exactly for nineteen new values of n , and by the Burger–Mynhardt bound (1) the largest gives

$$\gamma(Q_n) \leq \frac{1193}{2379}n + \mathcal{O}(1), \quad \frac{1193}{2379} \approx 0.5014712,$$

improving the recent coefficient $112/221 \approx 0.506787$ reported by Kong [8]. The resulting constructions are reported in Section 4.

2 Problem formulation and representation

2.1 Board, diagonals, and Type E sets

Let $n = 4k + 1$. We identify the board with a square of side n in the Cartesian plane, centred at the origin with sides parallel to the coordinate axes, and refer to squares by the coordinates of their centres. Thus, the square (x, y) lies in column x and row y . Rows and columns are collectively called *orthogonals*. An orthogonal is called *even* or *odd* according to the parity of its label. A square (x, y) is called *even-even* if both x and y are even, and *odd-odd* if both are odd.

Following [1], a set of queens is called a *Type E set* if all queens lie on even orthogonals and every even orthogonal contains at least one queen. Since there are $2k + 1$ even rows and $2k + 1$ even columns, a Type E set of cardinality $2k + 1$ contains exactly one queen in each even row and each even column. In particular, all of its queens lie on even-even squares. Every square lying in an even row or an even column is therefore dominated orthogonally. Hence only odd-odd squares can remain undominated, and these must be dominated diagonally.

It is convenient to compress the even orthogonals by a factor of two. Thus, a queen at full-board coordinates $(2x, 2y)$ is represented by the point (x, y) on the *reduced board*, whose row and column labels range from $-k$ to k . Unless stated otherwise, all coordinates and diagonal labels below refer to this reduced board.

Diagonals rising from left to right are called *d-diagonals* and are labelled by $d = y - x$, while diagonals falling from left to right are called *s-diagonals* and are labelled by $s = x + y$. For a Type E set, let i be the smallest positive integer such that at least one of the two diagonals $s = i$ and $s = -i$ is empty, and let j be defined analogously for the *d*-diagonals. Such a set is called an (i, j) -set.

We use the following result, stated as Theorem 2(a) of [1]. An (i, i) -set on Q_{4k+1} that contains a queen on every diagonal

$$s, d = 0, \pm 1, \pm 2, \dots, \pm(i - 1), \pm(i + 1), \pm(i + 3), \dots, \pm(2k - i - 1) \quad (3)$$

is dominating. Conversely, every dominating (i, i) -set D with $|D| = 2k + 1$ contains a queen on each of the diagonals in (3).

Here, as throughout, the diagonal labels in (3) and in the definition of an (i, j) -set are those of the reduced board. This is also the convention of [1], whose *d*- and *s*-labels range over $\{-2k, \dots, 2k\}$; a reduced-board label ℓ corresponds to the full-board diagonal offset 2ℓ . We state this explicitly because Östergård and Weakley [5] instead retain full-board labels and carry the factor two in their notation.

2.2 The parameter family

This work focuses on constructing centrally symmetric Type E (i, i) -sets of cardinality $2k + 1$. The centre square $(0, 0)$ is occupied, while the remaining $2k$ queens occur in k antipodal pairs: whenever (x, y) contains a queen, so does $(-x, -y)$.

We further restrict to even values of i , and write $i = e + 1$ and $u = k - e - 1$, so that e is odd. The positive diagonal labels required by (3) are then

$$R = \{1, 2, \dots, e\} \cup \{e + 2j : j = 1, \dots, u\}, \quad |R| = e + u = k - 1, \quad (4)$$

in each of the two diagonal directions.

Each noncentral antipodal pair $\{(x, y), (-x, -y)\}$ occupies the *d*-diagonals $\pm|y - x|$ and the *s*-diagonals $\pm|x + y|$. We restrict attention to pairs for which both of these magnitudes are nonzero. Thus each antipodal pair contributes one positive *d*-label and one positive *s*-label.

The k antipodal pairs therefore contribute k positive labels, counted with multiplicity, in each diagonal direction, whereas R contains only $k - 1$ required labels. Hence there is one

additional label in each direction. Denote these labels by a and b . The positive diagonal-label multisets are therefore

$$\mathbf{D} = R \uplus \{a\}, \quad \mathbf{S} = R \uplus \{b\}, \quad (5)$$

where $a, b > 0$, and each multiset has cardinality k . The additional labels need not be new: a or b may coincide with an element of R , in which case $\mathbf{D}$ or $\mathbf{S}$ contains a repeated label. However, to preserve $i = e + 1$ as the first empty diagonal distance, we require $a \neq i$ and $b \neq i$. Interchanging the d - and s -directions exchanges a and b , so we assume without loss of generality that $a \leq b$.

We emphasise that the centrally symmetric requirement, together with the restriction to antipodal pairs whose two diagonal magnitudes are both nonzero and the restriction to even i , defines a *proper subfamily* of the Type E (i, i) -sets of cardinality $2k + 1$. Half-turn symmetry has been used in much earlier work on queen domination, but it is known not always to be attainable, even when the required line sets are themselves symmetric [5]. Consequently, failure of the search to produce a set for a given k does not show that no Type E (i, i) -set of cardinality $2k + 1$ exists for that k . It shows only that none exists in this subfamily.

2.3 Pairing and coordinate coverage

For fixed parameters (k, e, a, b) , the multisets $\mathbf{D}$ and $\mathbf{S}$ are completely determined. We seek a bijection between their occurrences, written $\pi : \mathbf{D} \rightarrow \mathbf{S}$. For a pair (d, s) , with $d \in \mathbf{D}$ and $s = \pi(d) \in \mathbf{S}$, define

$$p = \frac{d + s}{2}, \quad q = \frac{|d - s|}{2}. \quad (6)$$

Solving $y - x = \pm d$ and $x + y = \pm s$ shows that a corresponding queen has coordinate magnitudes $\{|x|, |y|\} = \{p, q\}$. Not every pair (d, s) , however, is admissible in the family considered here. We call a pair *legal* if

$$d \equiv s \pmod{2}, \quad d \neq s, \quad d + s \leq 2k. \quad (7)$$

The first condition makes p and q integers, the second gives $q \geq 1$, and the third gives $p \leq k$. Since $d, s > 0$, we also have $p - q = \min\{d, s\} > 0$; in particular $p \neq q$, and every legal pair satisfies $1 \leq q < p \leq k$.

A feasible bijection must consist entirely of legal pairs and must also place the resulting queens in distinct reduced rows and columns. The next proposition shows that the latter requirement can be expressed solely in terms of the associated values p and q .

Proposition 1. *Let $\pi : \mathbf{D} \rightarrow \mathbf{S}$ be a bijection all of whose pairs are legal, and let (p_t, q_t) , $t = 1, \dots, k$, be given by (6). Suppose that every $v \in \{1, \dots, k\}$ occurs exactly twice in the multiset $\{p_1, q_1, \dots, p_k, q_k\}$. Then the pairs can be assigned coordinates so that, together with their antipodes and the centre, they form a centrally symmetric Type E (i, i) -dominating set of Q_{4k+1} with $i = e + 1$.*

Proof. Construct a multigraph G on vertex set $\{1, \dots, k\}$ with one edge $\{p_t, q_t\}$ for each t . Since $p_t \neq q_t$ for every t , as noted after (7), no edge is a loop. Since every value occurs exactly twice among the p_t and q_t , every vertex of G has degree two. Thus G is a disjoint union of cycles, with two parallel edges regarded as a cycle of length two.

Orient each cycle consistently. For the oriented edge $u \rightarrow v$ corresponding to pair t , assign $|x_t| = u$ and $|y_t| = v$. Every vertex then occurs exactly once as a tail and exactly once as a head. Consequently, the k representatives use each nonzero column magnitude and each nonzero row magnitude exactly once. Adding their antipodes gives exactly one queen in every nonzero reduced row and column, while the centre occupies row and column zero.

For each pair, $\{p_t + q_t, p_t - q_t\} = \{d_t, s_t\}$. Once $|x_t|$ and $|y_t|$ have been fixed by the cycle orientation, the two sign choices for x_t and y_t give $(|x_t + y_t|, |y_t - x_t|)$ equal to either

($p_t + q_t, p_t - q_t$) or ($p_t - q_t, p_t + q_t$); the sign choice is therefore independent of the cycle orientation, and suitable signs realize the prescribed d_t - and s_t -diagonal magnitudes. The resulting configuration therefore occupies every diagonal whose positive label belongs to R , in both diagonal directions. Since $i = e + 1 \notin R$ and $a, b \neq i$, the d - and s -diagonals at distance i remain empty. The resulting set is thus an (i, i) -set satisfying the diagonal condition (3), and hence is dominating by Theorem 2(a) of [1]. It is Type E by the row and column construction above and has cardinality $2k + 1$. $\square$

For later use, let $c(v)$ denote the number of occurrences of $v \in \{1, \dots, k\}$ in $\{p_1, q_1, \dots, p_k, q_k\}$, and define the *covering defect*

$$E(\pi) = \sum_{v=1}^k |c(v) - 2|. \quad (8)$$

Thus $E(\pi) = 0$ exactly when every coordinate magnitude occurs twice, as required in Proposition 1.

2.4 Orientation-free halved-label pairing representation

The preceding subsections assemble the Type E structure needed for the search and specialize ideas already present in the literature to the centrally symmetric family considered here. We now introduce the representation used in our computational search. Its main purpose is to remove several choices that would otherwise have to be represented explicitly during the search.

By (7), every legal pair consists of labels of the same parity. We therefore separate the odd and even labels and rescale the two classes. The key observation is that, after this transformation, the two labels of a pair need not be assigned to the d - and s -directions during the search. Instead, the search can be performed over unordered pairs, with the missing direction assignments reconstructed afterwards. As shown below, this loses no solutions within the parameter family considered here and leads directly to the constraint formulation used in our computational search.

2.4.1 Two halved universes

Put

$$m = \frac{e+1}{2}, \quad K = k - m. \quad (9)$$

Lemma 2. *The odd elements of R are exactly $1, 3, 5, \dots, 2K - 1$, and its even elements are exactly $2, 4, \dots, 2m - 2$.*

Proof. Since e is odd, $\{1, \dots, e\}$ contributes the odd values $1, 3, \dots, e$, of which there are m , and the even values $2, 4, \dots, e - 1$, of which there are $m - 1$. Again since e is odd, every $e + 2j$ is odd, contributing $e + 2, e + 4, \dots, e + 2u$. The odd elements of R are therefore all odd numbers from 1 to $e + 2u = 2k - e - 2 = 2K - 1$, and there are K of them. $\square$

A legal pair joins two labels of equal parity, so π splits into a bijection between the odd parts of $\mathbf{D}$ and $\mathbf{S}$ and a bijection between their even parts. In particular $a \equiv b \pmod{2}$, since otherwise those two odd parts have different sizes. Define the *halving map*

$$\mathbf{h}(\ell) = \begin{cases} (\ell - 1)/2, & \ell \text{ odd,} \\ \ell/2, & \ell \text{ even,} \end{cases} \quad (10)$$

and let κ be the number of pairs with odd labels, so that $\kappa = K + 1$ if a and b are odd and $\kappa = K$ if they are even. Applying $\mathbf{h}$ to the odd, respectively even, parts of $\mathbf{D} \uplus \mathbf{S}$ produces the two multisets

$$W_o = \{0, 0, 1, 1, \dots, K - 1, K - 1\} \uplus \Xi_o, \quad W_e = \{1, 1, 2, 2, \dots, m - 1, m - 1\} \uplus \Xi_e, \quad (11)$$

211 where $\Xi_o = \{\frac{a-1}{2}, \frac{b-1}{2}\}$ and $\Xi_e = \emptyset$ when a, b are odd, and $\Xi_o = \emptyset$, $\Xi_e = \{\frac{a}{2}, \frac{b}{2}\}$ when they are
 212 even. Thus $|W_o| = 2\kappa$ and $|W_e| = 2(k - \kappa)$. Both universes consist of consecutive integers, each
 213 repeated twice, together with at most two further elements.

214 **Proposition 3.** *Fix (k, e, a, b) with e odd and $a \equiv b \pmod{2}$. There exists a legal bijection*
 215 $\pi : \mathbf{D} \rightarrow \mathbf{S}$ *with $E(\pi) = 0$ if and only if there exist perfect matchings M_o of W_o and M_e of W_e*
 216 *into unordered pairs of distinct values such that, writing each pair with its smaller entry first,*
 217 *the multiset of emitted values*

$$\mathcal{V} = \biguplus_{\{X < Y\} \in M_o} \{Y - X, X + Y + 1\} \uplus \biguplus_{\{V < U\} \in M_e} \{U - V, U + V\} \quad (12)$$

218 *is equal to $\{1, 1, 2, 2, \dots, k, k\}$.*

219 *Proof.* Let (d, s) be a legal pair. If d and s are odd, write $d = 2X' + 1$ and $s = 2Y' + 1$
 220 with $X' = h(d)$, $Y' = h(s)$, and set $X = \min(X', Y')$, $Y = \max(X', Y')$. Then (6) gives
 221 $q = |d - s|/2 = Y - X$ and $p = (d + s)/2 = X + Y + 1$. If d and s are even, write $d = 2V'$,
 222 $s = 2U'$ and set $V = \min(V', U')$, $U = \max(V', U')$; then $q = U - V$ and $p = U + V$. In both
 223 cases $d \neq s$ is equivalent to the two halved values being distinct, and $d + s \leq 2k$ is equivalent
 224 to $p \leq k$, while $q \geq 1$ is automatic. Forgetting, for each pair, which of its two labels came
 225 from $\mathbf{D}$ leaves exactly a perfect matching of W_o and one of W_e , and $\mathcal{V}$ is precisely the multiset
 226 $\{p_1, q_1, \dots, p_k, q_k\}$; so $E(\pi) = 0$ says $\mathcal{V} = \{1, 1, \dots, k, k\}$.

227 Conversely, given such matchings, Lemma 4 below assigns one element of each pair to $\mathbf{D}$
 228 and the other to $\mathbf{S}$ with the prescribed multiplicities, that is, it produces a bijection π . Its pairs
 229 are legal and $E(\pi) = 0$ by the computation just made. $\square$

230 *Remark.* Three things are gained. First, both universes are runs of consecutive integers with
 231 multiplicity two plus at most two extra elements; the irregular arithmetic progression $\{e + 2j\}$
 232 of (4) has disappeared, and with it all parity bookkeeping. Second, the direction assignment has
 233 left the search, at no cost, by Lemma 4. Third, what remains is an exact-cover-type constraint
 234 problem: one variable per candidate pair, one equality per slot value, and one equality per value
 235 to be covered.

236 2.4.2 Recovering the diagonal directions

237 For $\tau \in \{o, e\}$, let $\mathbf{D}_\tau$ and $\mathbf{S}_\tau$ denote the submultisets of $\mathbf{D}$ and $\mathbf{S}$ consisting of labels of parity
 238 τ . Thus $W_\tau = h(\mathbf{D}_\tau) \uplus h(\mathbf{S}_\tau)$, where $W_o = W_o$ and $W_e = W_e$.

239 **Lemma 4.** *Let $\tau \in \{o, e\}$, and let M_τ be a perfect matching of W_τ into pairs of distinct*
 240 *values. Form the multigraph H_τ whose vertices are the values occurring in W_τ and whose*
 241 *edges are the pairs of M_τ . Then the edges of H_τ can be oriented so that, for every vertex v ,*
 242 *$\text{out}(v) = \text{mult}_{h(\mathbf{D}_\tau)}(v)$, and $\text{in}(v) = \text{mult}_{h(\mathbf{S}_\tau)}(v)$. Consequently, the two endpoints of every*
 243 *matched pair can be assigned consistently to the d - and s -directions. Such an orientation can*
 244 *be computed in time $O(|M_\tau|)$.*

245 *Proof.* Write $\delta_D(v) = \text{mult}_{h(\mathbf{D}_\tau)}(v)$ and $\delta_S(v) = \text{mult}_{h(\mathbf{S}_\tau)}(v)$. Since M_τ is a perfect matching
 246 of $h(\mathbf{D}_\tau) \uplus h(\mathbf{S}_\tau)$, we have $\deg_{H_\tau}(v) = \delta_D(v) + \delta_S(v)$ for every vertex v .

247 First suppose that τ is not the parity of a and b . Then the additional labels do not occur
 248 in $\mathbf{D}_\tau$ or $\mathbf{S}_\tau$, so these two multisets are identical. Hence $\delta_D(v) = \delta_S(v)$ for every v , and every
 249 vertex of H_τ has even degree. Orient each connected component along an Eulerian circuit.
 250 Then

$$\text{out}(v) = \text{in}(v) = \frac{\deg_{H_\tau}(v)}{2} = \delta_D(v) = \delta_S(v),$$

251 as required.

Now suppose that τ is the common parity of a and b . If $a = b$, then again $\mathbf{D}_\tau = \mathbf{S}_\tau$, and the same Eulerian argument applies.

It remains to consider $a \neq b$. Put $v_a = \mathbf{h}(a)$ and $v_b = \mathbf{h}(b)$. Since $\mathbf{h}$ is injective on labels of fixed parity, $v_a \neq v_b$. The multisets $\mathbf{D}_\tau$ and $\mathbf{S}_\tau$ agree except for the additional occurrence of a in $\mathbf{D}_\tau$ and of b in $\mathbf{S}_\tau$. Therefore $\delta_D(v_a) = \delta_S(v_a) + 1$ and $\delta_S(v_b) = \delta_D(v_b) + 1$, while $\delta_D(v) = \delta_S(v)$ at every other vertex. Thus v_a and v_b are the only vertices of odd degree in H_τ ; since every component of a multigraph contains an even number of vertices of odd degree, they lie in the same connected component.

Add an auxiliary edge ε joining v_a and v_b . All vertices of $H_\tau + \varepsilon$ then have even degree. Orient each component along an Eulerian circuit, choosing the orientation of the component containing ε so that $\varepsilon : v_b \rightarrow v_a$. After deleting ε , the indegree of v_a and the outdegree of v_b each decrease by one. Hence

$$\text{out}(v_a) - \text{in}(v_a) = 1, \quad \text{out}(v_b) - \text{in}(v_b) = -1,$$

while every other vertex remains balanced. Together with $\deg(v) = \delta_D(v) + \delta_S(v)$, this gives $\text{out}(v) = \delta_D(v)$ and $\text{in}(v) = \delta_S(v)$ for every v .

Eulerian circuits can be found using Hierholzer's algorithm in time linear in the number of edges, so the reconstruction takes $O(|M_\tau|)$ time. $\square$

Remark. Lemma 4 is a special case of a classical result on orientations with prescribed out-degrees, due to Hakimi [14]: a graph G admits an orientation with $d_G^+(v) = q(v)$ for every $v \in V(G)$ if and only if $\sum_{v \in V(G)} q(v) = |E(G)|$ and $e_G(S) \leq \sum_{v \in S} q(v)$ for every $S \subseteq V(G)$, where $e_G(S)$ is the number of edges with both ends in S ; see Frank and Gyárfás [15] for the version with both lower and upper bounds on the out-degrees. In the situation of Lemma 4 the prescribed out-degrees agree with $\deg_{H_\tau}(v)/2$ at all but at most two vertices, so the required orientation is the standard balanced orientation obtained from an Eulerian circuit. We include the short direct proof above because it is constructive and yields the linear-time reconstruction used in Section 3.3.

Lemma 4 is what makes the orientation-free formulation exact: the omitted d/s assignments can always be recovered without search. In fact, this recovery requires only linear-time Eulerian traversals of the matching components. Combined with the cycle orientation of Proposition 1, this yields the reconstruction procedure of Section 3.3, which contains no search and just takes a few milliseconds on the largest board reported here.

2.5 Admissible parameter choices

The preceding sections developed the representation used for the search and showed that a solution of the pairing problem can be reconstructed into a Type E configuration without further search. The search still depends on the parameters (k, e, a, b) , however, and not every choice of these parameters can correspond to a feasible configuration.

It is therefore useful to eliminate some impossible parameter choices before the search procedure is run. Necessary arithmetic conditions of this kind have long been used in computational searches for special classes of queen configurations. In fact, the use of such conditions to restrict a small parameter space before a combinatorial search is due to Östergård and Weakley [5]. Here we draw on such conditions from the literature and, where appropriate, specialize them to the centrally symmetric Type E family considered above. For convenience, put

$$T = \sum_{\ell \in \mathbf{D}} \ell + \sum_{\ell \in \mathbf{S}} \ell = 2(K^2 + m(m-1)) + a + b. \quad (13)$$

Also, for a multiset W with $|W| = 2P$, let $\text{ext}(W)$ denote the sum of its P smallest elements. Since a perfect matching of W contains P pairs, the sum obtained by taking the smaller element

from each pair can never be less than $\text{ext}(W)$. For later use, define the corresponding bounds in the original diagonal-label units by

$$\text{ext}_\ell(W_o) = 2 \text{ext}(W_o) + \kappa, \quad \text{ext}_\ell(W_e) = 2 \text{ext}(W_e).$$

With these definitions in place, we can now state the necessary conditions used to filter the parameter choices.

Condition (F1) is Weakley's Parallelogram Law [16], also stated in [5, Theorem 4] and [6, Section 3.1], and recorded in a publicly accessible form as equation (23) of [17], specialized to the present family. We record it here in our notation for completeness. The identity underlying (F3), $\sum_t \max(d_t, s_t) = k(k+1)$, also appears in [8, Lemma 5]; the form of (F3) used here is a weaker consequence of that identity.

Proposition 5. *If the parameters (k, e, a, b) admit a configuration in the family considered here, then the following conditions hold.*

(F1)

$$a^2 + b^2 = N(k, e),$$

where

$$N(k, e) = \frac{2}{3} \left(k(k+1)(2k+1) - K(4K^2 - 1) - 2m(m-1)(2m-1) \right).$$

(F2)

$$a \equiv b \pmod{2}, \quad 1 \leq a \leq b \leq 2k-1, \quad a \neq e+1, \quad b \neq e+1, \quad \kappa \equiv 0 \pmod{2}.$$

(F3)

$$S(k, e, a, b) := T - k(k+1) - \text{ext}_\ell(W_o) - \text{ext}_\ell(W_e) \geq 0,$$

Moreover, $S(k, e, a, b)$ is even.

Proof. For (F1), (6) gives

$$p^2 + q^2 = \frac{d^2 + s^2}{2}.$$

Since every value $1, \dots, k$ occurs exactly twice among the p - and q -values,

$$2 \sum_{v=1}^k v^2 = \frac{1}{2} \left(\sum_{\ell \in \mathbf{D}} \ell^2 + \sum_{\ell \in \mathbf{S}} \ell^2 \right) = \frac{1}{2} \left(2 \sum_{\ell \in R} \ell^2 + a^2 + b^2 \right).$$

Using Lemma 2 to evaluate the sum over R and rearranging gives (F1).

For (F2), the equal-parity condition was already observed in Section 2.4, the bounds on a and b follow from the range of positive diagonal labels, and $a \leq b$ is the symmetry convention fixed in Section 2.2; the conditions $a \neq e+1$ and $b \neq e+1$ were imposed there in order to preserve $i = e+1$ as the first empty diagonal distance. To see that κ is even, note that an odd pair $\{X < Y\}$ emits $Y - X$ and $X + Y + 1$, exactly one of which is odd, whereas an even pair emits either zero or two odd values. Because every value in $\{1, \dots, k\}$ is emitted twice, the total number of odd emitted values is even, and hence so is κ .

For (F3), exact coverage gives

$$\sum_t \max(d_t, s_t) = \sum_t (p_t + q_t) = k(k+1),$$

and hence

$$\sum_t \min(d_t, s_t) = T - k(k+1).$$

320 If an odd matched pair has halved entries $X < Y$, then $\min(d, s) = 2X + 1$, whereas for an
 321 even pair $V < U$, $\min(d, s) = 2V$. Thus, if $\mu(P)$ denotes the smaller halved entry of a pair P ,

$$T - k(k + 1) = 2 \sum_P \mu(P) + \kappa.$$

322 For any perfect matching,

$$\sum_P \mu(P) \geq \text{ext}(W_o) + \text{ext}(W_e).$$

323 Returning to the original label units therefore gives

$$T - k(k + 1) \geq \text{ext}_\ell(W_o) + \text{ext}_\ell(W_e),$$

324 which is (F3). Moreover,

$$S(k, e, a, b) = 2 \left(\sum_P \mu(P) - \text{ext}(W_o) - \text{ext}(W_e) \right),$$

325 so $S(k, e, a, b)$ is even. □

326 Condition (F1) is particularly effective: for fixed (k, e) it reduces the choice of (a, b) to rep-
 327 resentations of $N(k, e)$ as a sum of two squares. Conditions (F2) and (F3) provide inexpensive
 328 additional filters. Any parameter choice that violates one of these conditions can be discarded,
 329 whereas a choice that passes all three must still be tested by the search procedure. In our
 330 experiments for $k \leq 400$, conditions (F1) and (F2) reduced the parameter space to 183 admis-
 331 sible quadruples, and (F3) reduced this further to 101, spanning 76 values of k . The complete
 332 filtering step took well under a second.

333 3 Computational search

334 For a fixed value of k , the search proceeds in two stages. We first enumerate the parameter
 335 quadruples (k, e, a, b) satisfying the restrictions of Section 2.2 and the necessary conditions of
 336 Proposition 5, ordered by decreasing S . The surviving quadruples are then considered in this
 337 order. For each one, the halved universes W_o and W_e are constructed and the pairing problem
 338 of Proposition 3 is solved as a constraint-satisfaction problem, using several randomized runs
 339 before moving to the next quadruple.

340 In particular, the search is performed directly in the orientation-free representation. Thus
 341 the solver does not assign labels to the d - and s -directions, nor does it assign row and column
 342 roles. It only determines how the occurrences in W_o and W_e are paired. A feasible solution must
 343 use every occurrence exactly once and must emit every coordinate magnitude $1, \dots, k$ exactly
 344 twice. By Proposition 3, these two requirements are sufficient, and the omitted orientations can
 345 be recovered afterwards without further search. Algorithm 1 summarizes this search procedure.

Algorithm 1 SEARCH(k, t, r)

Input: board parameter k , per-run time limit t , and r runs per parameter quadruple

Output: set $\mathcal{C}$ of verified certificates found for the admissible parameter quadruples

```
1:  $\mathcal{C} \leftarrow \emptyset$ 
2: enumerate the parameter quadruples  $(k, e, a, b)$  satisfying the restrictions of Section 2.2 and
   Prop. 5, ordered by decreasing  $S$ 
3: for all surviving quadruples  $(k, e, a, b)$  do
4:   construct the CP-SAT model of Section 3.1
5:   for  $\sigma \leftarrow 0$  to  $r - 1$  do
6:     set the time limit to  $t$  and random seed to  $\sigma$ 
7:     if  $\sigma > 0$  then
8:       enable randomized search with random-variable pool size 5
9:     end if
10:    solve the model
11:    if feasible then
12:      reconstruct and verify the configuration as in Section 3.3
13:      add the verified certificate to  $\mathcal{C}$ 
14:      break ▷ move to the next parameter quadruple
15:    else if infeasible then
16:      break ▷ the quadruple is refuted
17:    end if
18:  end for
19: end for
20: return  $\mathcal{C}$ 
```

3.1 Constraint encoding

This section describes how the pairing problem of Proposition 3 is encoded as a constraint-satisfaction problem. Consider one of the two halved universes. For each unordered pair $\{X, Y\}$ of distinct values with $X < Y$, the emitted values are $Y - X$, and $X + Y + \theta$, where $\theta = 1$ for W_o and $\theta = 0$ for W_e . We retain the pair as a candidate only if both emitted values lie in $\{1, \dots, k\}$.

Let $\nu_\tau(X)$ denote the multiplicity of X in the universe corresponding to $\tau \in \{o, e\}$. For every retained candidate pair $\{X, Y\}$, introduce an integer variable

$$z_{XY}^\tau \in \{0, \dots, \min(\nu_\tau(X), \nu_\tau(Y))\},$$

whose value is the number of times that pair is used in the matching.

Two families of constraints are imposed. First, every occurrence of every halved value must belong to exactly one matched pair:

$$\sum_{\substack{\{U, V\} \text{ candidate} \\ X \in \{U, V\}}} z_{UV}^\tau = \nu_\tau(X) \quad \text{for every value } X \text{ in half } \tau. \quad (14)$$

These constraints enforce a perfect matching of each multiset.

Second, the two matchings together must emit every coordinate magnitude exactly twice. Let $m_v^\tau(U, V) \in \{0, 1, 2\}$ denote the number of times the candidate pair $\{U, V\}$ emits the value v . We impose

$$\sum_{\tau \in \{o, e\}} \sum_{\{U, V\} \text{ candidate}} m_v^\tau(U, V) z_{UV}^\tau = 2, \quad v = 1, \dots, k. \quad (15)$$

The coefficient $m_v^\tau(U, V)$ is two when both outputs of a candidate pair are equal to v .

The first family of constraints consumes exactly the available multiplicity of every value in W_o and W_e , while the second enforces the exact coordinate coverage required by Proposition 3. Consequently, every feasible assignment determines precisely the two matchings needed for reconstruction. Note that there is no objective function and the model is a pure feasibility problem. Also note that the variables are integer rather than binary. We solve the resulting models with CP-SAT [18].

3.2 Randomized restarts

Previous computational work on restricted queen configurations has reported substantial variation in search time [6]. We observe similar behaviour with CP-SAT, with a strong dependence on the random seed. Across four representative boards, using 8–10 runs per board under a 60 s time limit, success rates ranged from 5 out of 10 to 7 out of 10, while successful runs took between 2.0 and 48.5 s.

Such variation motivates restart strategies in randomized SAT and constraint-satisfaction search [19]. We therefore perform several runs for each admissible parameter quadruple, varying the CP-SAT random seed. For nonzero seeds, randomized search is additionally enabled with a random-variable pool size of five. The appropriate restart length, however, depends on the instance. For example, splitting a 60 s budget into four 15 s runs increases the estimated success probability from 0.70 to 0.97 at $k = 193$, but decreases it from 0.50 to 0.34 at $k = 241$, where no observed successful run took less than 11.9 s. These estimates are based on at most ten runs per configuration and are indicative only; we therefore use multiple seeds as a means of diversifying the search, without claiming that the chosen restart schedule is optimal.

Search difficulty is also non-monotone in k , so the available computational budget is distributed across both parameter choices and solver seeds rather than concentrated on a single run.

3.3 Reconstruction and verification

A feasible solver assignment specifies the multiplicity of every selected unordered pair and hence determines perfect matchings M_o and M_e . These matchings are converted into queen coordinates in two deterministic steps.

First, Lemma 4 orients the matching edges so that their endpoints acquire the required d - and s -roles. Second, the emitted values form the 2-regular multigraph of Proposition 1; orienting each of its cycles assigns the two coordinate magnitudes of every antipodal pair to the row and column directions. Appropriate signs then recover the reduced coordinates, and adding their antipodes together with the centre gives the full set of $2k + 1$ queens. Neither reconstruction step requires search.

The resulting certificate is finally checked by a separate verifier, implemented independently of the search code. The verifier confirms the cardinality and board coordinates, verifies that every even row and column contains exactly one queen, checks the required d - and s -diagonal conditions for the claimed Type E (i, i) -set, and directly verifies domination of the board by testing all n^2 squares. Thus the reported constructions do not depend on trusting the internal state of the constraint solver.

4 Main Results

The computations were performed using two Intel Xeon CPU cores at 2.80 GHz. The per-run time limit and the number of random seeds were varied across instances, reflecting the substantial variation in search difficulty described in Section 3.2. The time spent on parameter

filtering, model construction, and verification was negligible compared with the solver runtime. Thus, only the solve time of the successful run that produced each reported certificate is given.

Table 1 lists the nineteen Type E (i, i) -sets obtained. In every case, $i = e + 1$ and the set has cardinality $2k + 1$. All constructions were obtained using the same implementation and parameter filter, and were verified as described in Section 3.3. Constructions were also found for smaller values of k , including $k = 28$ and $k = 43$ (see Table 2); Table 1 lists only those with $k > 55$, since $k = 55$ is the largest value for which a set of this kind was previously known [6]. The final column reports the asymptotic coefficient from (1). For the largest instance, $n = 1585$, substituting $k = 396$ into (1) gives

$$\gamma(Q_n) \leq \frac{1193}{2379}n + \mathcal{O}(1), \quad \frac{1193}{2379} \approx 0.5014712,$$

for all sufficiently large n . This improves the recent asymptotic coefficient $112/221 \approx 0.506787$ reported by Kong [8], and also the coefficient $30/59 \approx 0.5084746$ of Neuhaus [6, Cor. 3.24].

Independently of (1), the constructions of Table 1 determine nineteen new values of the domination number.

Corollary 6. *For each of the nineteen values of n listed in Table 1,*

$$\gamma(Q_n) = \frac{n+1}{2}.$$

Proof. Each certificate is a dominating set of Q_{4k+1} of cardinality $2k+1$, so $\gamma(Q_{4k+1}) \leq 2k+1$. Weakley’s lower bound $\gamma(Q_{4k+1}) \geq 2k+1$ [3] gives equality, and $2k+1 = (n+1)/2$. $\square$

The largest n for which a value of this kind was previously known is $n = 221$ [6, 8]. All nineteen values in Table 1 are therefore new, and extend the range of known exact values of $\gamma(Q_n)$ from $n = 221$ to $n = 1585$.

Certificates and verification. The certificates for the constructions reported in Table 1, together with a Python verifier, are publicly available at <https://github.com/TahaRostami/Gamma/tree/upper-bound>.

5 Effect of the orientation-free representation

Lemma 4 shows that the representation of Section 2.4 loses no solutions, but not that it is easier to search. To test this we encoded the bijection of Section 2.3 directly, using one integer variable for each legal pair (d, s) and the same coordinate-coverage constraints as in Section 3.1. In this formulation, each label retains its assignment to the d - or s -direction. We refer to this model as the *oriented* formulation and to the model of Section 3.1 as the *orientation-free* formulation. The comparison was performed using the same solver configuration in both cases: one worker, a time limit of 55 s per run, and seeds 0–4 as described in Section 3.2.

As shown in Table 2, the advantage of the orientation-free formulation becomes clear as k grows. For $k = 100$ and $k = 148$, none of the oriented runs succeeds within the time limit, whereas the orientation-free formulation solves all five seeds.

To separate the effect of removing the direction assignments from that of halving the labels, we also tested unordered pairs of the original labels. This formulation performed similarly to the halved model, indicating that the main gain comes from removing the d/s assignments rather than from the halving itself. The halving is nevertheless retained because it is what makes the bound ext_ℓ of Section 2.5, and hence condition (F3), available in closed form.

This comparison uses five instances and five seeds per instance and is intended to isolate the effect of the representation within a fixed solver and encoding, rather than to establish a general performance claim; in particular, the conclusion is specific to the CP-SAT formulation used here and does not necessarily imply a comparable advantage for other search paradigms.

n	k	e	u	a	b	S	$\gamma(Q_n)$	solve (s)	$(3k+5)/(6k+3)$
249	62	25	36	19*	79*	2	125	0.3	191/375 $\approx$ 0.5093333
285	71	29	41	74	74	32	143	2.9	218/429 $\approx$ 0.5081585
401	100	43	56	164	186	36	201	0.4	305/603 $\approx$ 0.5058043
421	105	45	59	170	190	40	211	1.3	320/633 $\approx$ 0.5055292
429	107	45	61	134	166	52	215	1.4	326/645 $\approx$ 0.5054264
509	127	53	73	116	184	56	255	0.8	386/765 $\approx$ 0.5045752
521	130	55	74	146	238	52	261	0.9	395/783 $\approx$ 0.5044700
561	140	59	80	176	222	68	281	3.1	425/843 $\approx$ 0.5041518
569	142	59	82	120	170	60	285	47.0	431/855 $\approx$ 0.5040936
593	148	63	84	200	284	56	297	3.7	449/891 $\approx$ 0.5039282
773	193	81	111	236	250	98	387	172.8	584/1161 $\approx$ 0.5030146
865	216	91	124	140	398	52	433	19.0	653/1299 $\approx$ 0.5026944
885	221	93	127	238	342	104	443	33.8	668/1329 $\approx$ 0.5026336
917	229	97	131	302	412	98	459	7.2	692/1377 $\approx$ 0.5025418
965	241	101	139	220	312	108	483	103.1	104/207 $\approx$ 0.5024155
1209	302	127	174	348	392	152	605	16.8	911/1815 $\approx$ 0.5019284
1221	305	129	175	446	488	146	611	591.0	920/1833 $\approx$ 0.5019094
1321	330	139	190	360	490	160	661	548.0	995/1983 $\approx$ 0.5017650
1585	396	167	228	534	534	204	793	437.0	1193/2379 $\approx$ 0.5014712

Table 1: Verified Type E (i, i) -sets, with $i = e + 1$. In every case the dominating set has cardinality $2k + 1$, so that $\gamma(Q_n) = 2k + 1 = (n + 1)/2$ by Corollary 6. The solve column reports the CP-SAT solve time of the successful run that produced the reported certificate. The final column gives the asymptotic coefficient $(3k + 5)/(6k + 3)$. *: for $k = 62$, the free labels coincide with labels already present in R , so $\mathbf{D}$ and $\mathbf{S}$ are proper multisets (Section 2.2).

6 Conclusion

In this paper, we presented a computational method for constructing members of a centrally symmetric subfamily of Type E (i, i) -dominating sets of Q_{4k+1} with cardinality $2k + 1$. The main idea is to reformulate the construction problem as a pairing problem on diagonal labels. By separating the labels according to parity and halving them, several choices that would otherwise be represented explicitly during the search can be omitted and recovered deterministically afterwards. In particular, the diagonal-direction assignments and the row/column roles need not be searched directly.

We further restrict the admissible parameter choices using necessary arithmetic conditions and solve the resulting pairing instances by constraint programming with randomized restarts. Using this approach, we obtained nineteen verified constructions for larger values of k . These determine $\gamma(Q_n)$ exactly for nineteen new values of n between 249 and 1585 (Corollary 6), extending the range of known exact values from $n = 221$. The largest construction has $k = 396$ on Q_{1585} and, through the Burger–Mynhardt bound, gives

$$\gamma(Q_n) \leq \frac{1193}{2379}n + \mathcal{O}(1) \approx 0.5014712n + \mathcal{O}(1)$$

for all sufficiently large n , improving the previous best known asymptotic upper bound.

More broadly, the results show that removing unnecessary choices from the search representation can make substantially larger structured queen-domination constructions computationally accessible. An interesting direction for future work is to determine whether the regularity exposed by the pairing formulation can be exploited to obtain still larger constructions or eventually an explicit infinite family; the nineteen matchings underlying Table 1 are available in the repository cited above and may be a useful starting point.

k	oriented			orientation-free		
	vars	solved	median (s)	vars	solved	median (s)
28	375	5/5	0.0	190	5/5	0.0
43	907	5/5	16.2	462	5/5	0.0
62	1872	1/5	—	936	5/5	0.1
100	4939	0/5	—	2482	5/5	1.0
148	10863	0/5	—	5450	5/5	4.8

Table 2: Comparison of the oriented and orientation-free formulations using five seeds and a time limit of 55 s per run. Medians are computed over successful runs. “—” indicates that fewer than half of the runs succeeded.

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